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of Exeter

TM & NLP (COMM040)

Dr. Rodrigo Wilkens

Intended Learning Outcomes



- Understand the strengths, weaknesses and biases of different NLP methods
- Clean, parse, transform and process large collections of text documents
- Know about recent advances and applications of NLP methods, their impact and real world applications.
- Communicate the pros and cons and trade-offs of different approaches to text analysis

Outline

- Introduction to CA
- 1966 – now
- CA Evaluation
- Perspectives

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Introduction to CA

- Speech acts:

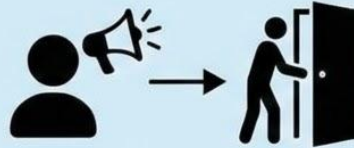
PERFORMATIVE ACTS



DOING THINGS WITH WORDS

I promise, I apologize,
I sentence, I hereby name

DIRECTIVES



INFLUENCE ACTION

Commands, Requests, Suggestions

CONSTATIVES



STATE OF AFFAIRS

Facts, True/False, Descriptions

COMMISSIVES



FUTURE COMMITMENT

Promises, Offers, Threats

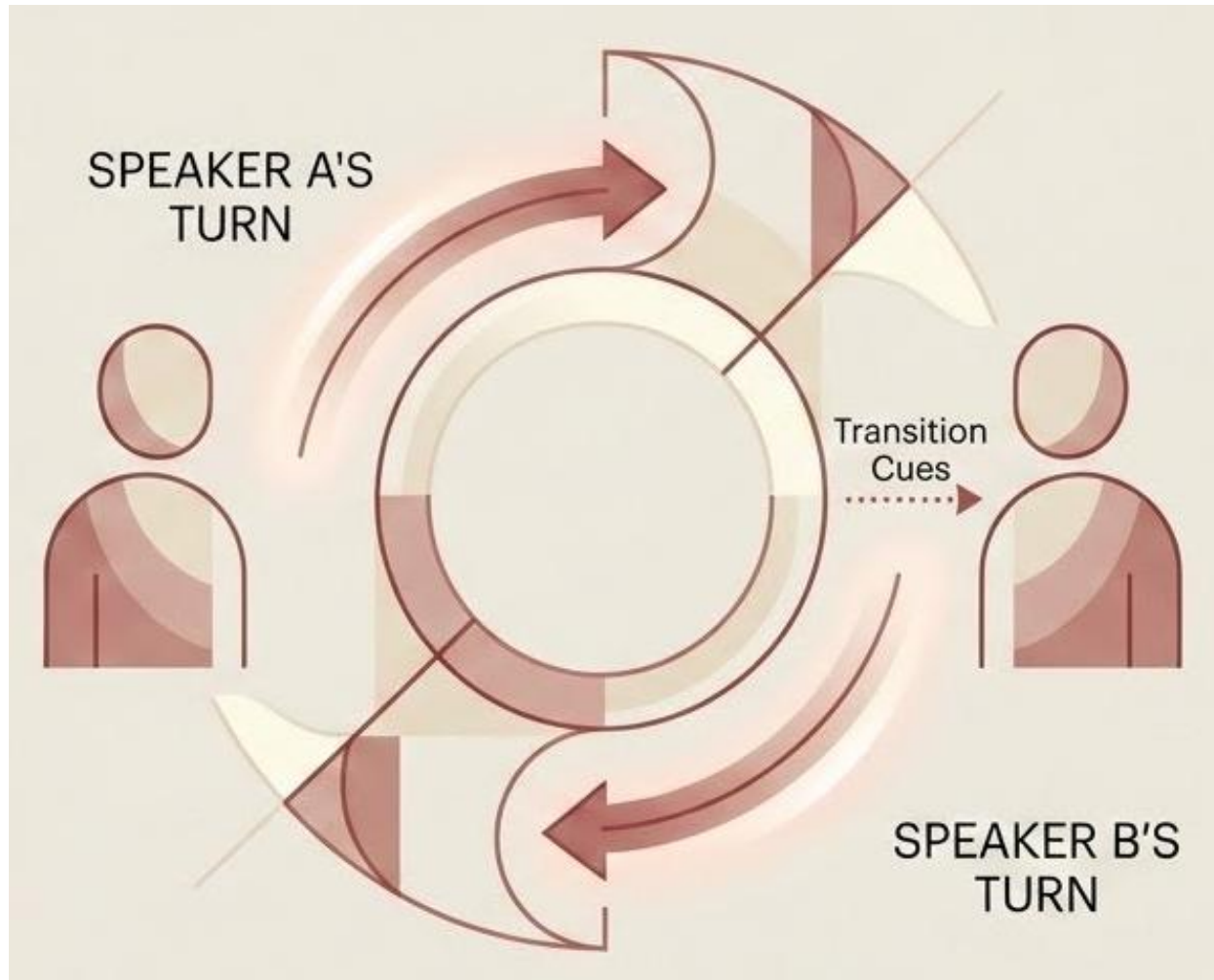
ACKNOWLEDGEMENTS



SIGNAL ENGAGEMENT

Hearing, Understanding, Validation

Introduction to CA



- **Turn:**
 - the blocks (phrases, words or clauses) that make up a contribution to discourse.
- **Turn-taking:**
 - The organisation in conversation and discourse where participants speak one at a time in alternating turns.

Introduction to CA

- Real conversation:

- Silence:

- Pause, gap, lapse

- Repair

- Adjacency pairs

- Q&A, Offer-Acceptance/Refusal and Compliment-Response

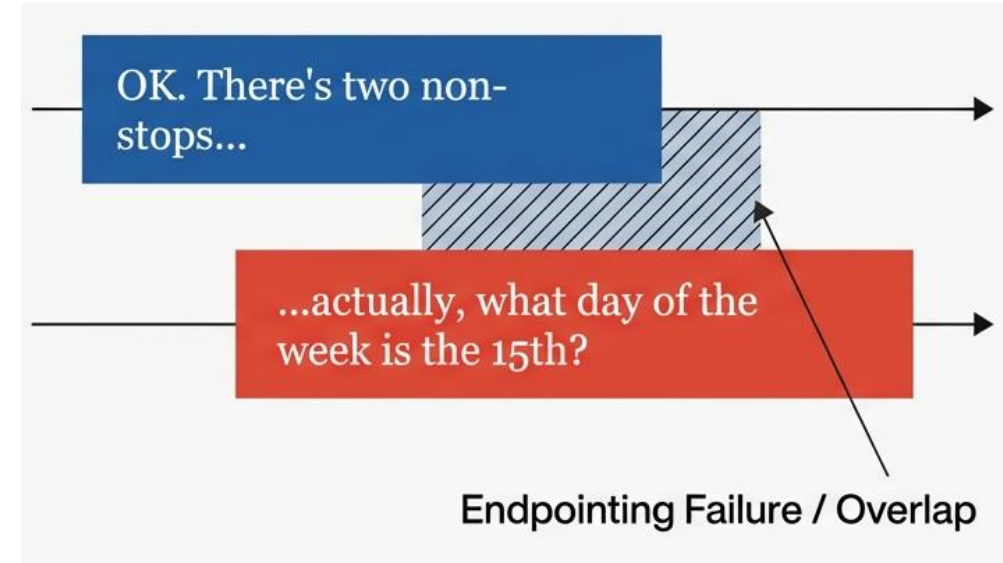
- Action formation

- assessments, storytelling, questions, and complaints

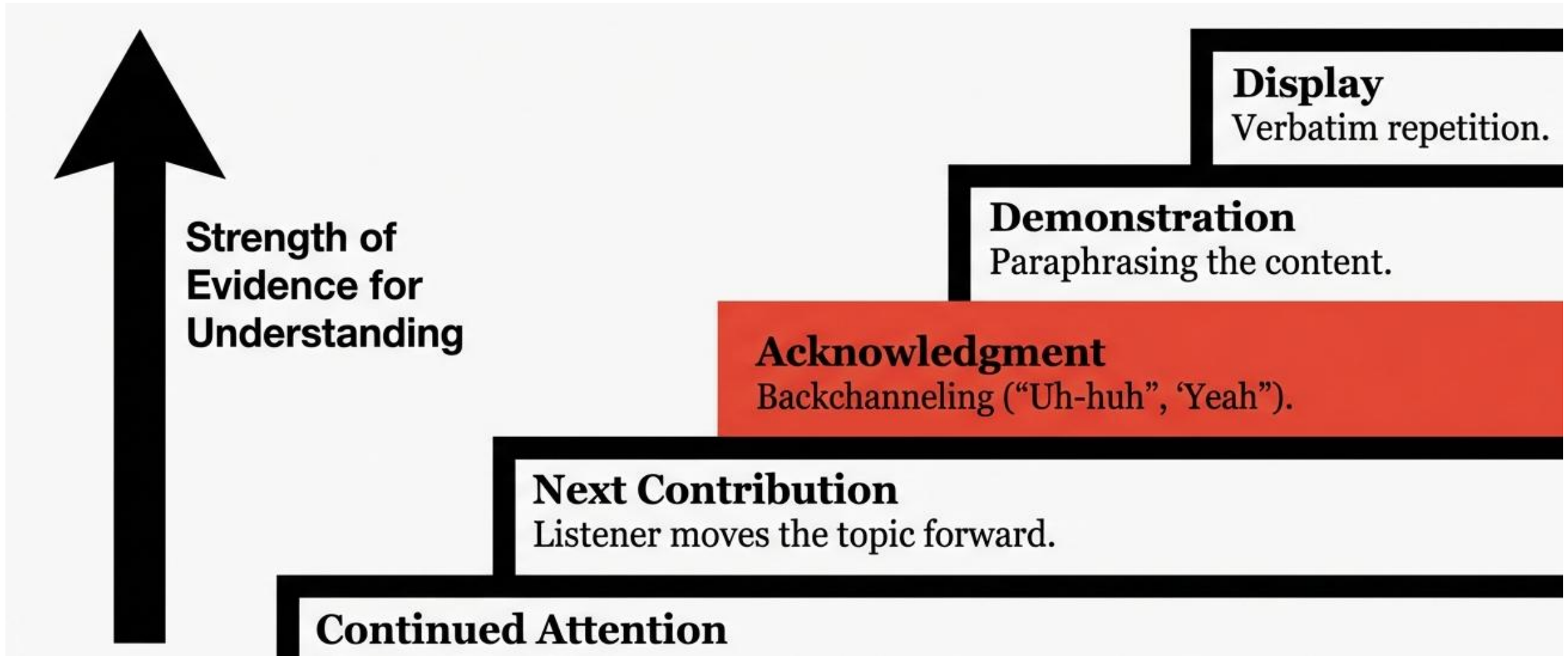
- Sequence expansion

- more than a single adjacency pair

I would like a turkey sandwich, please. (pair 1 start)
White or wholegrain? (pair 2 start)
Wholegrain. (pair 2 end)
Okay. (pair 1 end)



Introduction to CA



Introduction to CA



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Conversational Agent



Advanced NLU & Context: Understands complex intent, context, and nuance across multi-turn dialogues.



Stateful & Persistent: Remembers past interactions and user preferences for personalized experiences.



Goal-Oriented & Proactive: Capable of complex task completion, problem-solving, and initiating actions.



Examples: Siri, Alexa, Google Assistant, advanced enterprise virtual assistants.



Chatbot



Rule-Based or Simple NLP: Relies on predefined rules, keyword matching, or basic pattern recognition.



Stateless or Limited Context: Often treats each interaction independently; limited memory of prior turns.



Reactive & Informational: Primarily handles FAQs, simple queries, and guided, scripted flows.



Examples: FAQ bots, basic ordering bots, scripted customer service interfaces.

Outline

- Introduction to CA
- **1966 – now**
- CA Evaluation
- Perspectives

1966 – now

- Rule-based
 - Eliza, Parry, Alice...
- Frame-based
 - Gus, Siri, Woebot...
- LLM-based
 - ChatGPT, BlenderBot 3

1966 - now

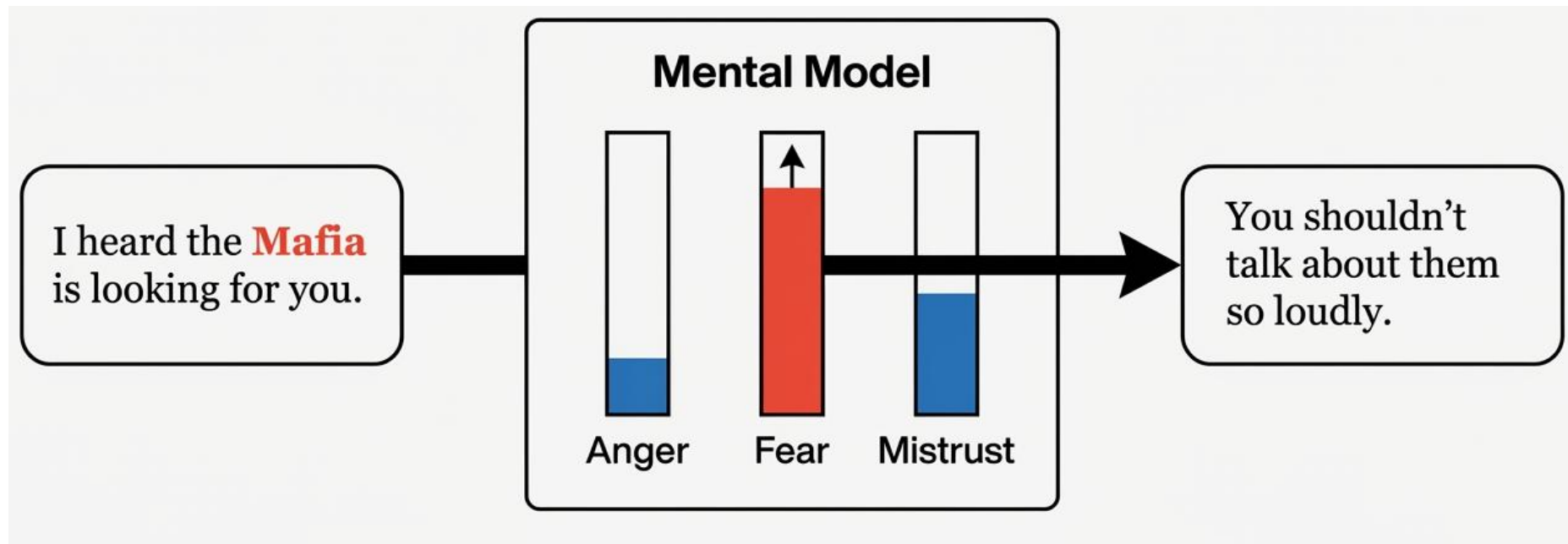
- ELIZA (1966)
 - Rules based on Regular Expression
 - Pairs of input-output
 - Rogerian Mirror Therapy

INPUT: "My boyfriend made me come here."
↓
SCAN: Keyword "boyfriend" -> Family/Relationship Category.
↓
TRANSFORM RULE: "My <X> made me..." -> "Your <X> made you..."
↓
OUTPUT: "Your boyfriend made you come here?"

> Men are all alike.
IN WHAT WAY
> They're always bugging us about something or other.
CAN YOU THINK OF A SPECIFIC EXAMPLE
> Well, my boyfriend made me come here.
YOUR BOYFRIEND MADE YOU COME HERE
> He says I'm depressed much of the time.
I AM SORRY TO HEAR YOU ARE DEPRESSED
> It's true. I am unhappy.
DO YOU THINK COMING HERE WILL HELP YOU NOT TO BE UNHAPPY
> I need some help, that much seems certain.
WHAT WOULD IT MEAN TO YOU IF YOU GOT SOME HELP
> Perhaps I could learn to get along with my mother.
TELL ME MORE ABOUT YOUR FAMILY
> My mother takes care of me.
WHO ELSE IN YOUR FAMILY TAKES CARE OF YOU
> My father.
YOUR FATHER
> You are like my father in some ways.
WHAT RESEMBLANCE DO YOU SEE
> You are not very aggressive but I think you don't want me to notice that.
WHAT MAKES YOU THINK I AM NOT VERY AGGRESSIVE
> You don't argue with me.
WHY DO YOU THINK I DON'T ARGUE WITH YOU
> You are afraid of me.
DOES IT PLEASE YOU TO BELIEVE I AM AFRAID OF YOU
> My father is afraid of everybody.
WHAT ELSE COMES TO MIND WHEN YOU THINK OF YOUR FATHER
> Bullies.
DOES THAT HAVE ANYTHING TO DO WITH THE FACT THAT YOUR BOYFRIEND MADE YOU COME HERE

1966 – now

- PARRY (1968)
 - Input words trigger internal state variables



1966 – now

- Alice (1995)



A.L.I.C.E. (Artificial Linguistic Internet Computer Entity) is a free software chatbot created in AIML (Artificial Intelligence Markup Language), an open, minimalist, stimulus-response language for creating bot personalities like A.L.I.C.E.

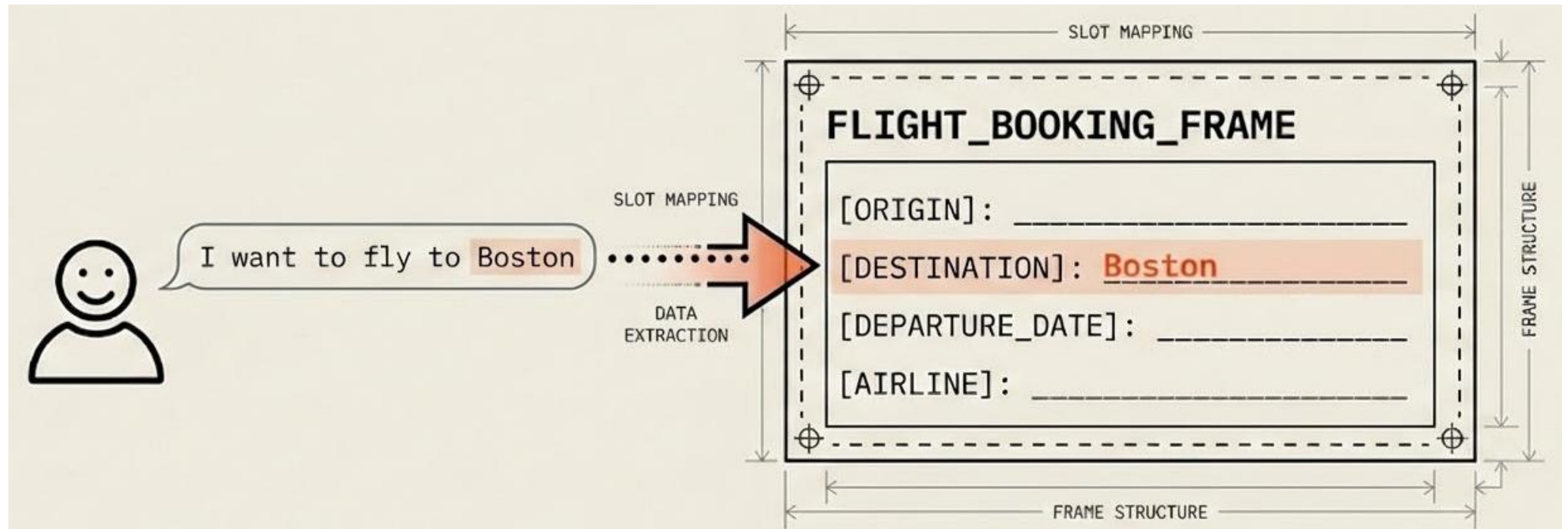
- Artificial Intelligence Markup Language (AIML)

```
<category>
  <pattern>WHAT IS YOUR NAME</pattern>
  <template><![CDATA[My name is <bot name="name"/>.]]></template>
</category>
<category>
  <pattern>WHAT ARE YOU CALLED</pattern>
  <template>
    <srai>what is your name</srai>
  </template>
</category>
```



1966 – now

- GUS (1977) and Siri (2011)
 1. Input is parsed and mapped to specific entities
 2. A frame is identified



1966 – now

- GUS (1977) and Siri (2011)



1966 – now

- Woebot (2017)
 - Keyword Matching
 - Rule-based policies
 - Internal variables

"It sounds like you're feeling {emotion}."

- ML



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If you're happy, I'm happy 😊

Happy days!



Hey can I ask you something?

Go for it

Is there anything you'd like to achieve for yourself in the next couple of weeks?



Like a personal goal?

No thanks

Great!

What would you like to achieve in the next couple of weeks?

Write it below... ✎



1966 – now

- Woebot (2017)
 - Dialogue Graph
 - Finite-State/Dialogue-State Machine
 - Belief state

State S0 (Greeting)

- State S1 (Mood check)
- State S2 (Cognitive distortion detection)
- State S3 (CBT intervention)
- State S4 (Reflection)
- State S5 (Closure)

1966 – now

- Woebot (2017)
 - Dialogue Graph
 - Finite-State/Dialogue-State Machine

State S0 (Greeting)

- State S1 (Mood check)
- State S2 (Cognitive distortion detection)
- State S3 (CBT intervention)
- State S4 (Reflection)
- State S5 (Closure)

- ML Intent Classification

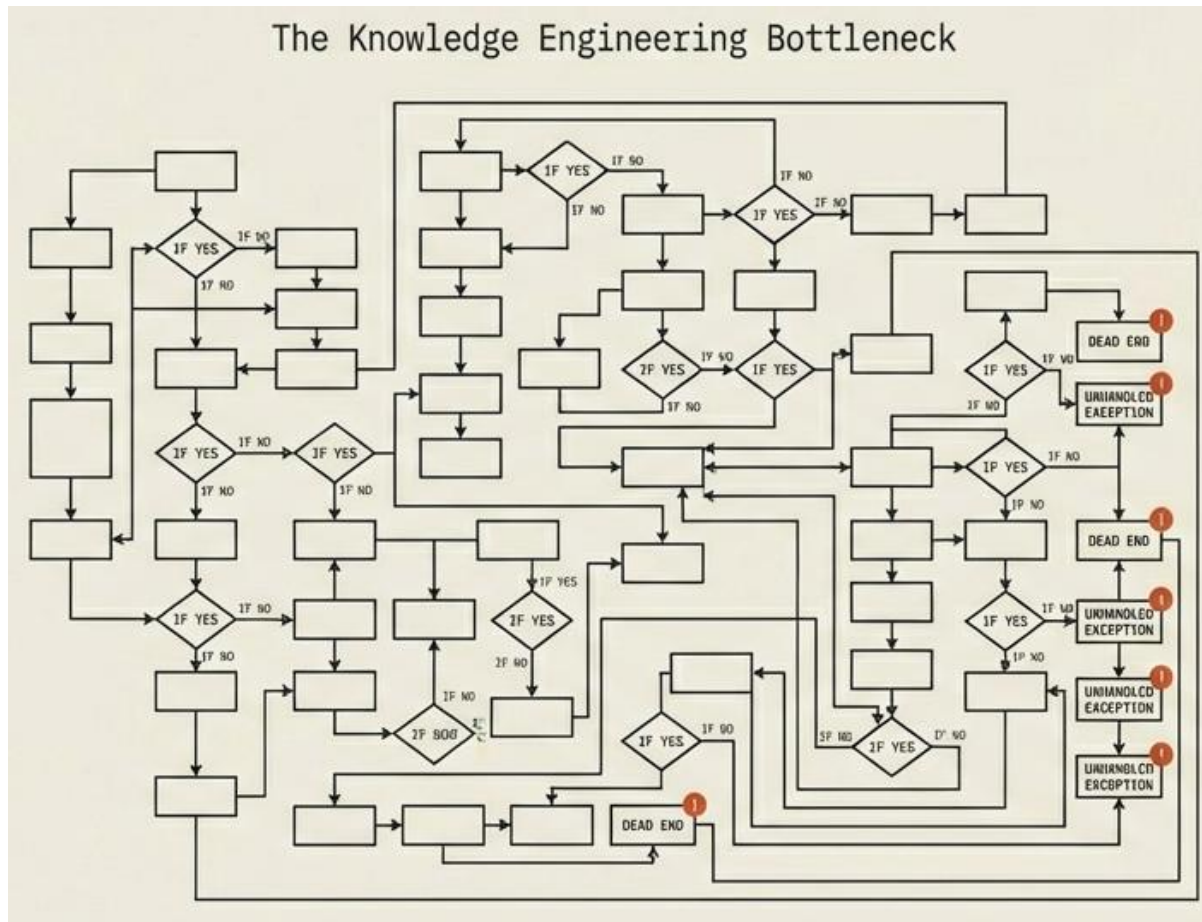
```
Intent: cognitive_distortion_self_blame  
Confidence: 0.87
```

- Entity Extraction (NER)

```
Event: exam  
Emotion: failure  
Time reference: yesterday
```

1966 - now

Exploring Representation of Knowledge



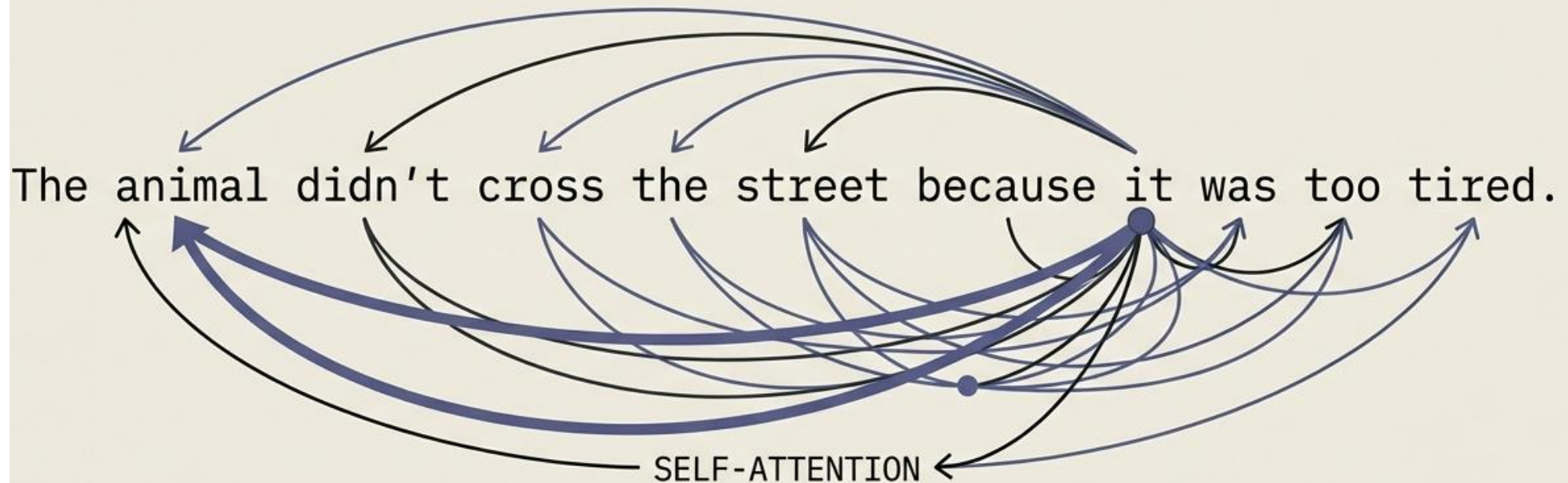
1966 – now

• LLM: ChatGPT

The Transformer: Processing entire context windows simultaneously.

Pre-training: Learning from internet-scale data (1 Trillion+ parameters).

Fine-tuning: RLHF alignment for instruction following.



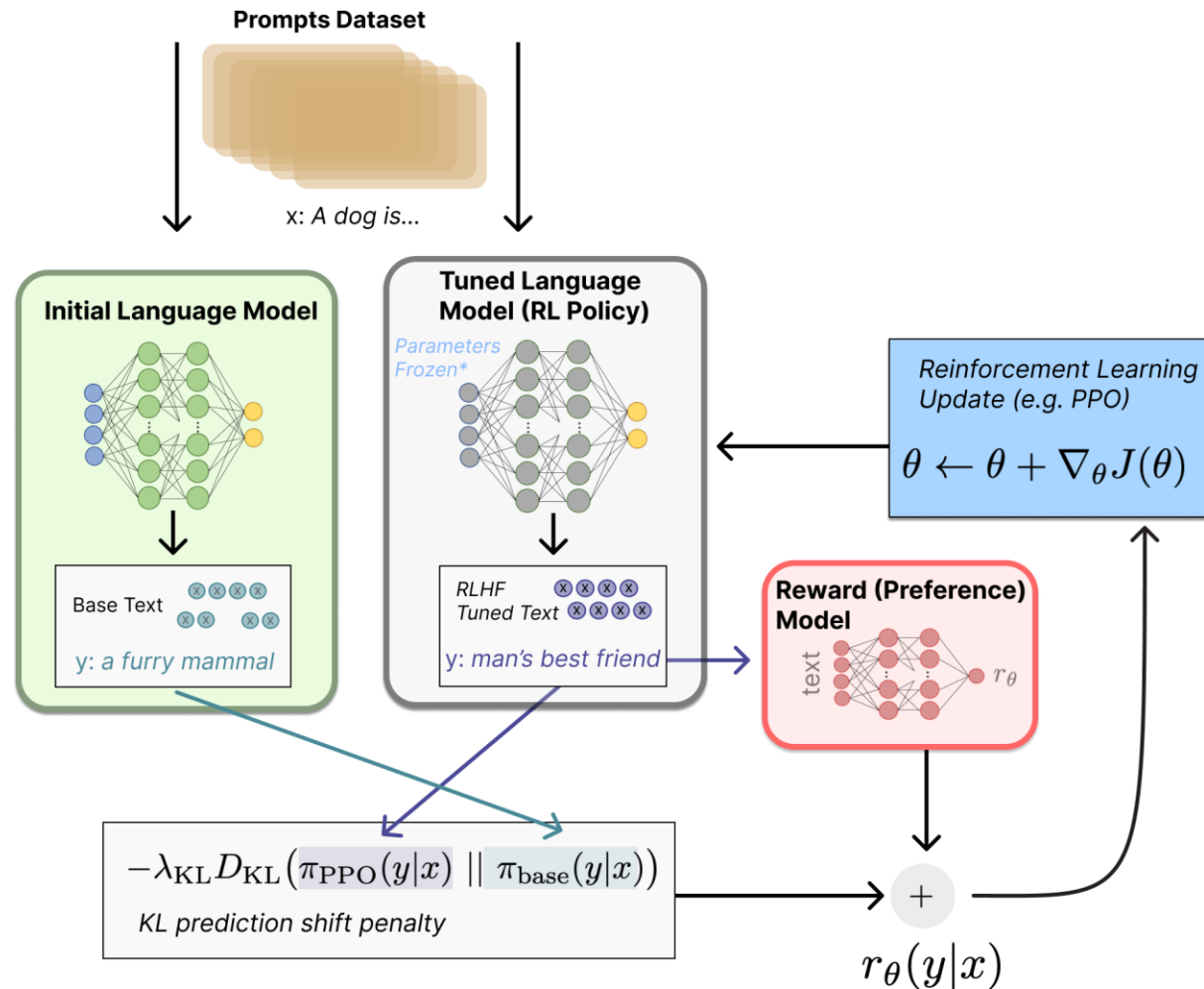
The model resolves the pronoun 'it' by attending to all words, with the strongest connection to 'animal'.

1966 – now

- LLM + RLHF
 - Turn-level adequacy
 - Conversational style
 - Politeness
 - Responsiveness

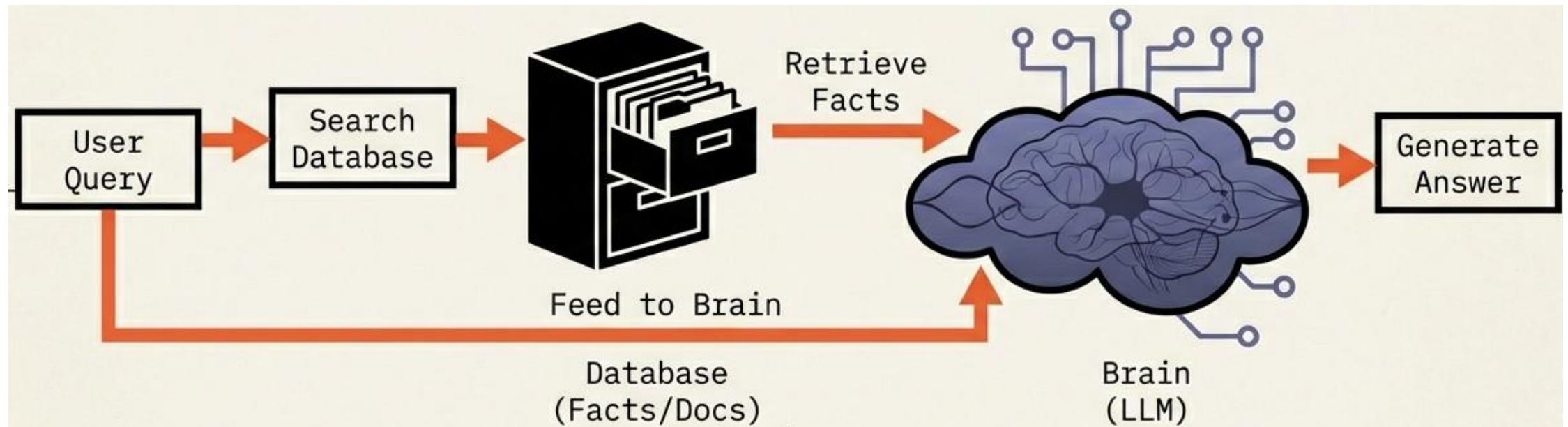
1966 - now

- LLM + RLHF



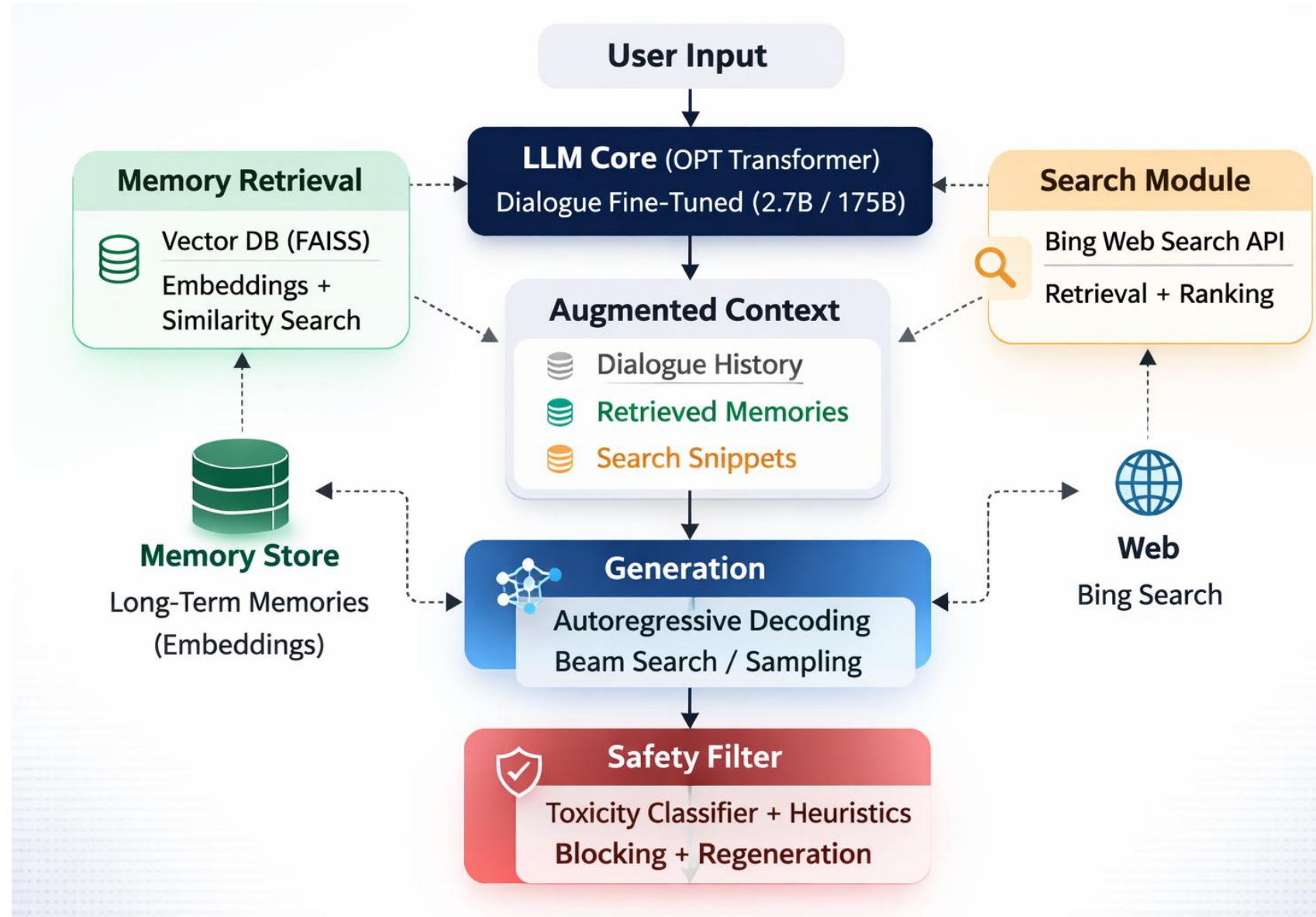
1966 - now

- LLM
 - Grounding the Parrot (RAG)

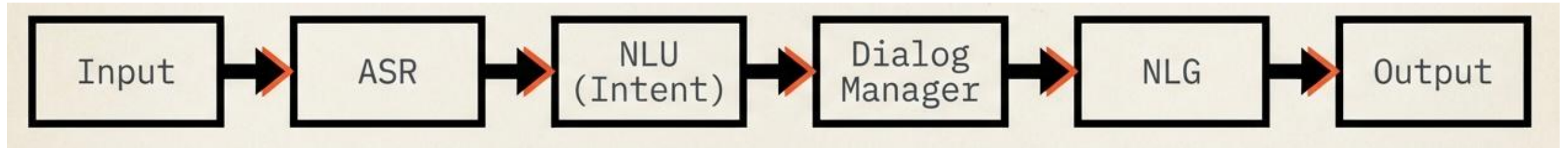


1966 – now

- LLM
 - BlenderBot 3



1966 - now



1966 – now

Era	Metaphor	Key Strength	Key Weakness
Pattern Matching	The Mirror	Robustness	No Intelligence
Frame-Based	The Bureaucrat	Precision	Rigidity
Neural/LLM	The Improviser	Creativity	Truthfulness

1966 – now

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1966 – now

- menti



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Outline

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- 1966 – now
- **CA Evaluation**
- Perspectives

CA Evaluation

- Frame-based approach

Make an appointment with Chris at 10:30 in Gates 104.

Slot	Filler
PERSON	Chris
TIME	11:30 a.m.
ROOM	Gates 104

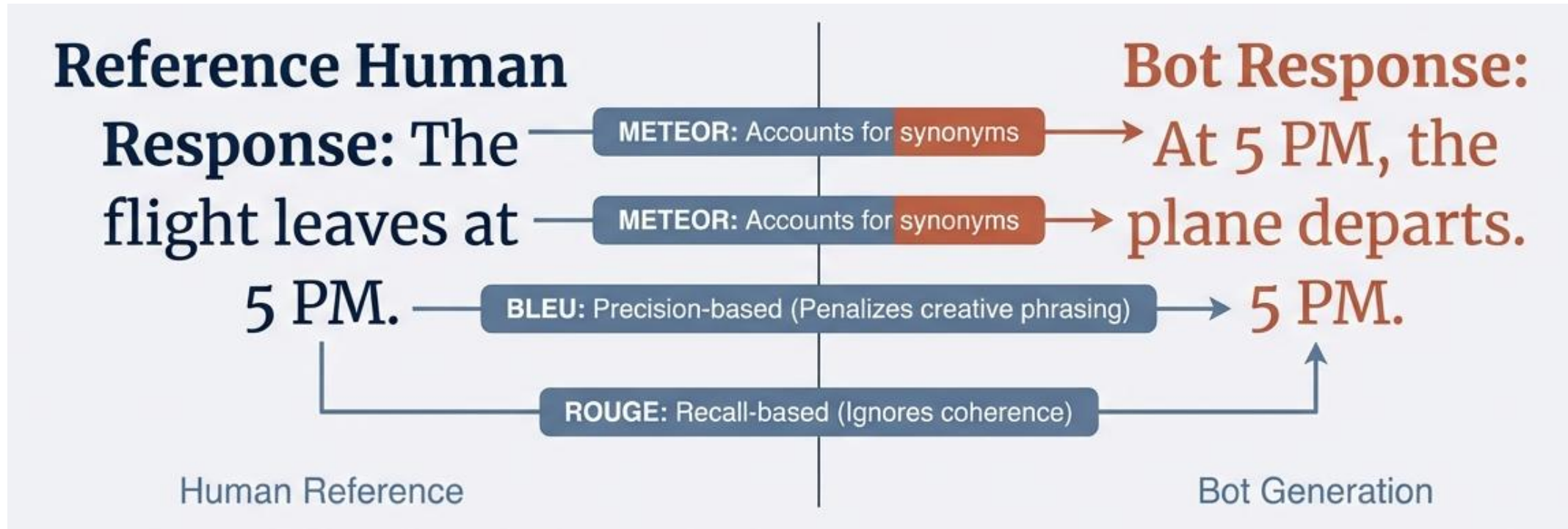
Slot Error Rate: 1/3 (33%) ↗

Task Success: FAIL
(Wrong time added)

F1 Score: Harmonic mean of
Precision and Recall

CA Evaluation

- NLG



CA Evaluation



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Speech Acts



Constatives: Stating facts.
Committing the speaker to something's
being the case.



Directives: Asking/Ordering.
Attempts by the speaker to get the
addressee to do something.



Commissives: Committing to action.
Committing the speaker to some future
course of action.



Acknowledgments: Confirming.
Expressing the speaker's attitude regarding
the hearer with respect to some social action.

Adjacency Pair
(Question/Answer)

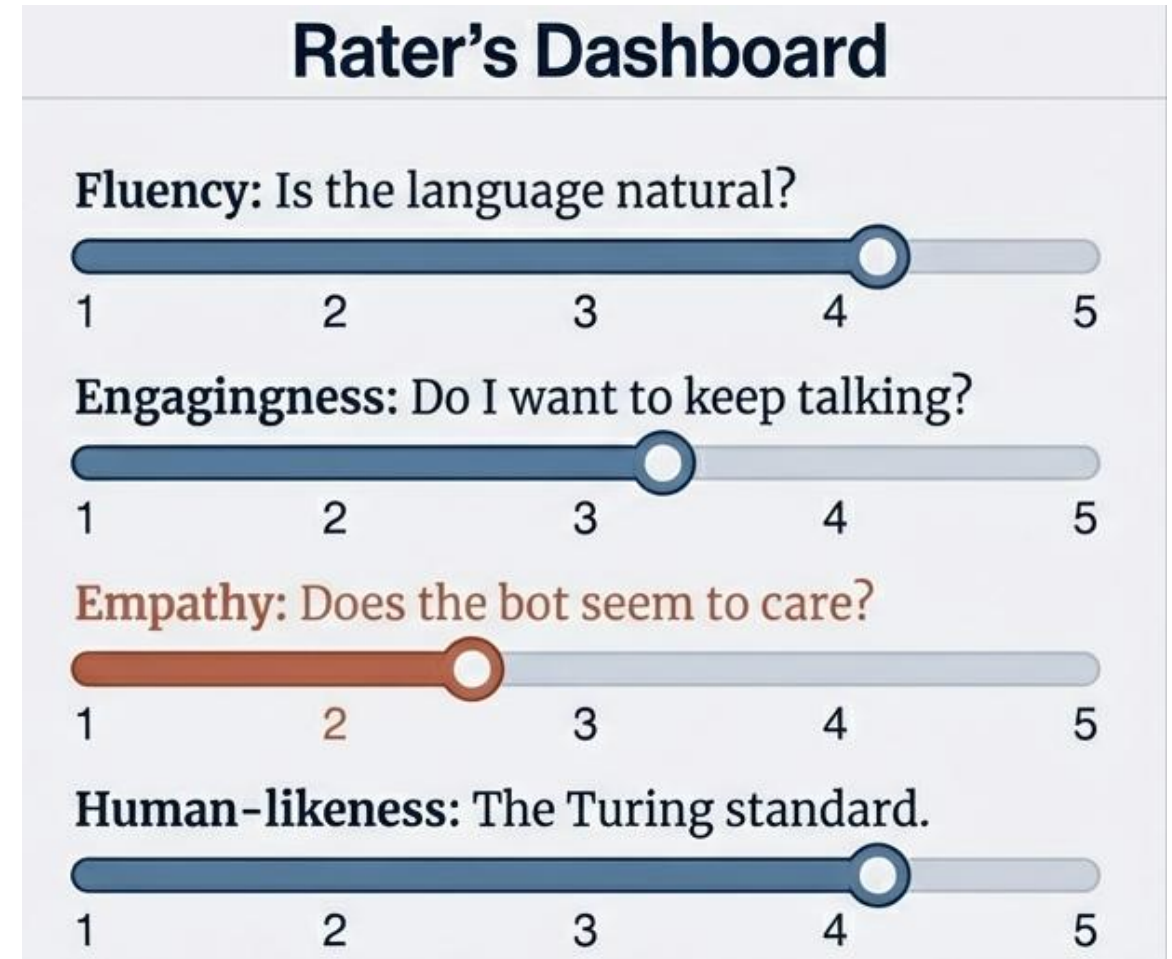
Endpointing Challenge
(Pause vs. Finish)

Grounding
(Acknowledgment)



CA Evaluation

- Human Evaluation and
- Domain Evaluation

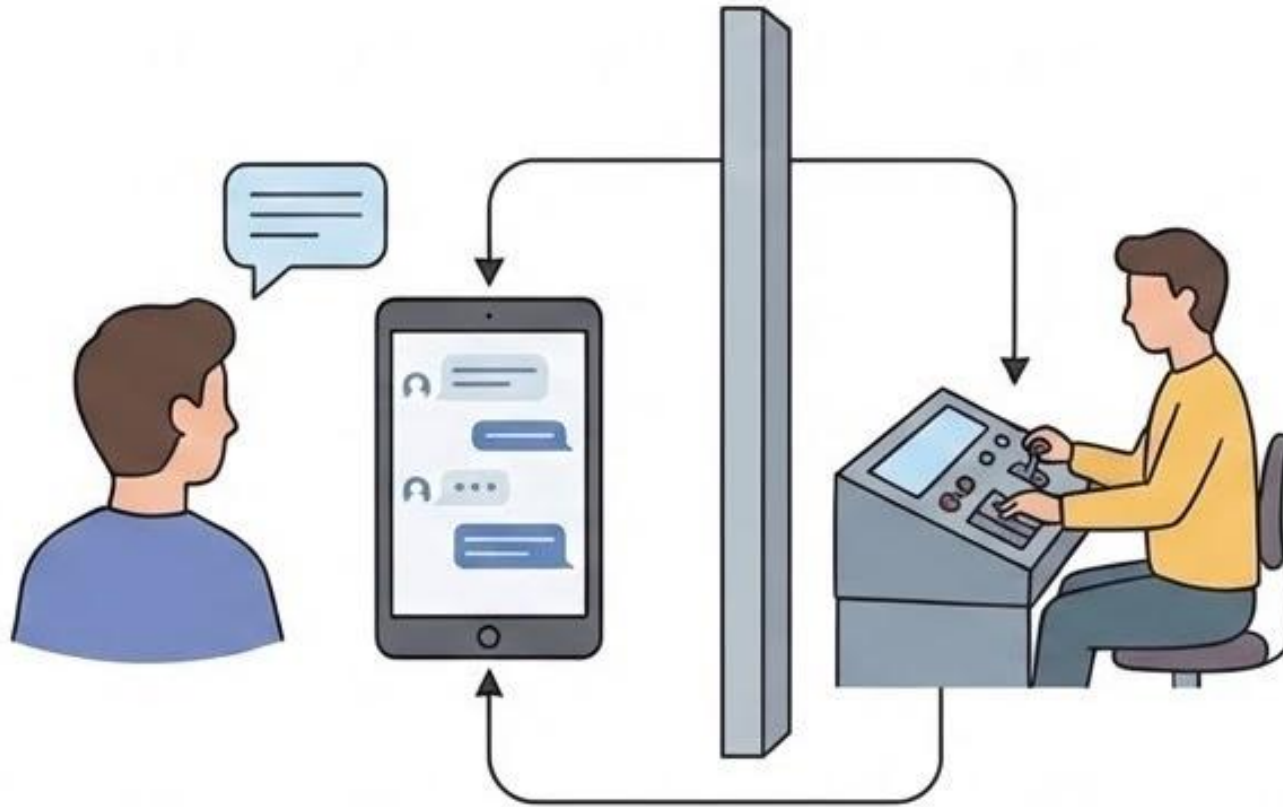


CA Evaluation

- Iteratively test the design on users
 - A/B testing:
 - Productions
 - Red Teaming
 - Adversarial test
 - Wizard of Oz
 - ...

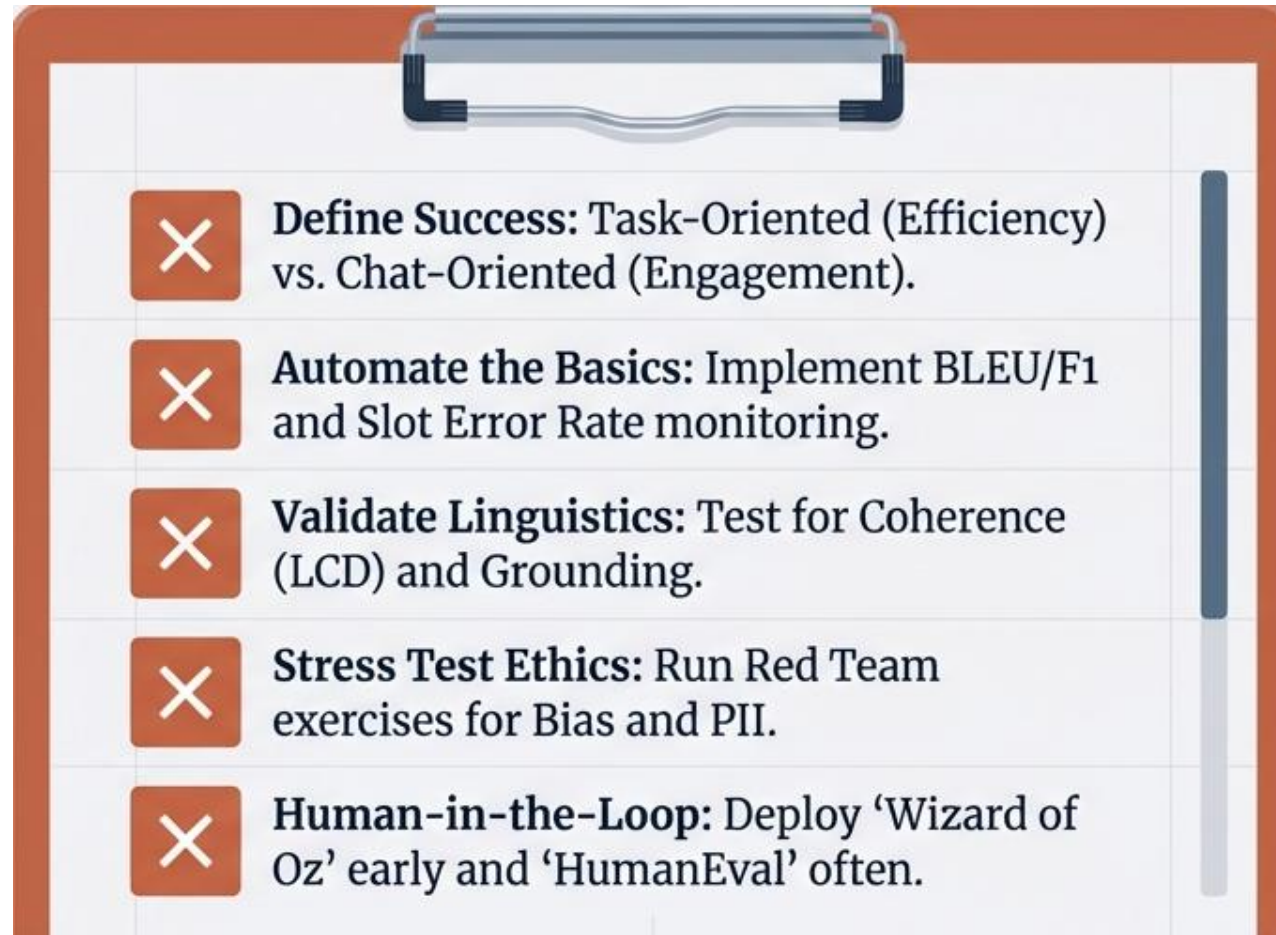
CA Evaluation

- Wizard of Oz



CA Evaluation

- Evaluation Checklist



Outline

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Perspectives

• Personas

Steve
Age: 51

Getting up in the morning is such an effort. I just don't see the point. Just tackling the stairs makes me feel worn out and out of puff.



His COPD has really been getting him down lately.
He's worried about his job as he's had to take so much time off and is getting out less and less.

Character traits:

Living circumstances:

Attitude towards technology:

Attitude towards own health:

Sources of support (friends and/or family):

Hobbies/Interests:

Christine
Age: 43

It's been quite a shock. I didn't expect to be told I had chronic lung disease at my age. There's a lot to take in and think about.



Has been having problems with her breathing and exercise tolerance for a while. She has recently attended her GP and has been given a diagnosis of COPD.

Character traits:

Living circumstances:

Attitude towards technology:

Attitude towards own health:

Sources of support (friends and/or family):

Hobbies/Interests:

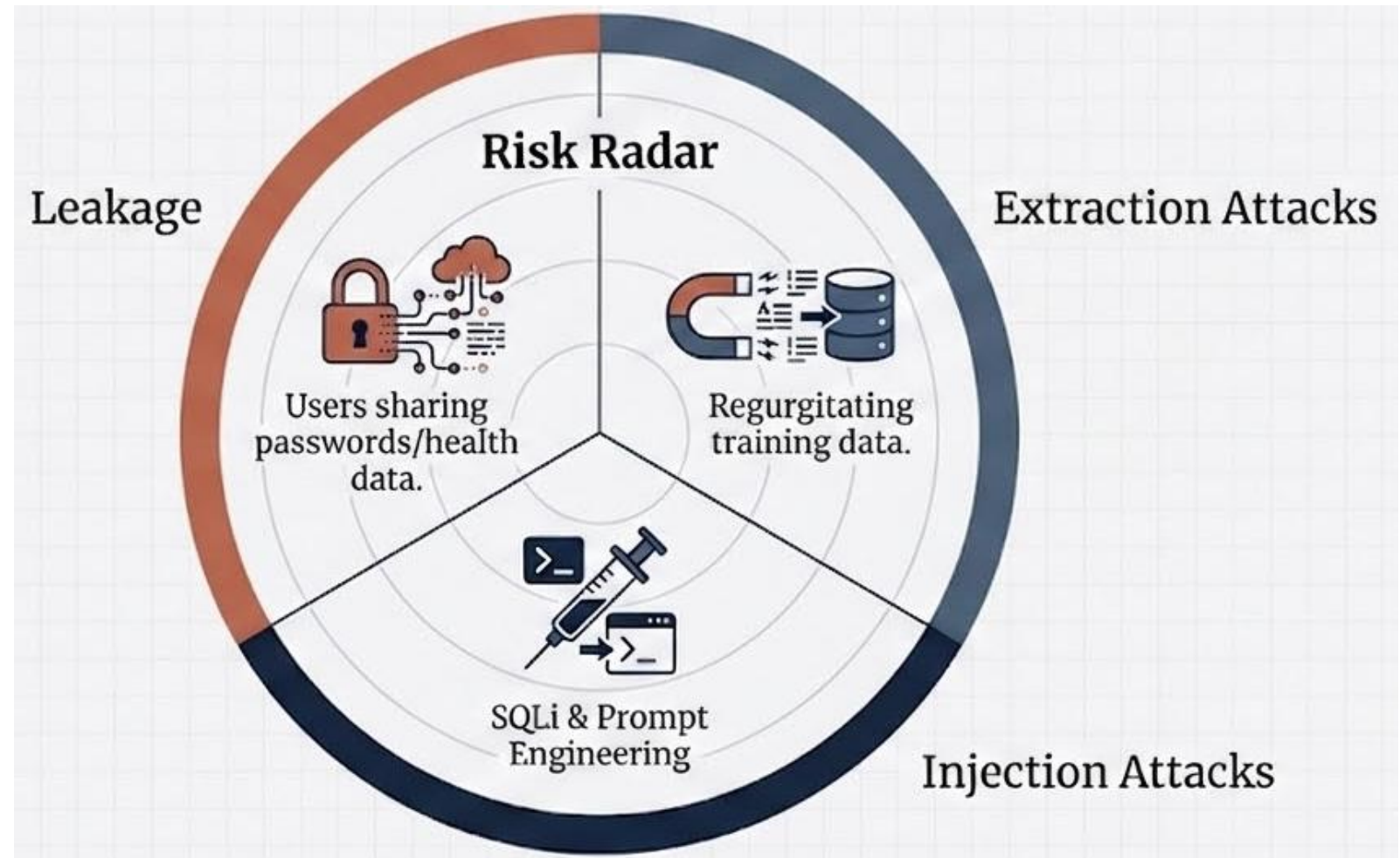
Perspectives

- Bias:
 - Garbage In, Garbage Out.
 - Response to harassment

Assistant	Response
Siri	I'd blush if I could.
Alexa	Well, thanks for the feedback.
Cortana	Well, that's not going to get us anywhere.
Google Home	My apologies, I don't understand.

Perspectives

- Red teaming



Perspectives

- Response: “I don’t know”

Perspectives

- Ethical and Societal Risks
 - Anthropomorphism
 - Emotional dependency
 - AI companionship
 - Data logging & consent
 - Workplace automation



Perspectives

- Research Frontiers
 1. Multimodal conversational agents
 2. Embodied dialogue
 3. Self-reflective agents
 4. Alignment & controllability
 5. Cultural adaptation

Exercise

- You are part of the evaluation team of a conversational agent. The system has been deployed with a defined persona. Your task is to design the test criteria.
- **Persona A:** *The Supportive Mental Health Coach*
 - Calm, empathetic tone
 - Non-judgmental
 - Encourages reflection
 - Avoids giving medical advice
 - De-escalates distress
- **Persona B:** *The Efficient Legal Assistant*
 - Formal tone
 - Precise language
 - Risk-aware
 - Avoids speculation
 - Does not provide definitive legal advice

Exercise

- **Persona A:** *The Supportive Mental Health Coach*
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 - **Persona B:** *The Efficient Legal Assistant*
 - Formal tone
 - Precise language
 - Risk-aware
 - Avoids speculation
 - Does not provide definitive legal advice
1. What elements must be tested to verify if this persona is consistent?
 2. How would you operationalise one of these dimensions?

Conclusion

- By the end of this session, students will be able to:
 - Compare rule-based, frame-based, and LLM-based conversational agents in control, flexibility, and reliability.
 - Apply intent classification and entity extraction concepts to structured dialogue management.
 - Explain how transformers and grounding approaches reshaped conversational agent architectures.
 - Evaluate conversational agents using slot error rate, task success, human evaluation, and bias considerations.